Physics 5313: Electromagnetic Theory 2 (Fall 2026)

Lecture 03: Reflection and Refraction

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Example: Faraday Rotation

Faraday Rotation During Propagation For propagation along the z-axis, a medium supports left circular polarization with index of refraction n_L and right circular polarization with index of refraction n_R. If a plane wave propagating through this medium has $\mathbf{E}(z=0,t) = \hat{\mathbf{x}}E\exp(-i\omega t)$, find the values of z where the wave is linearly polarized along the y-axis.

$$\vec{E}(\vec{x},t) = \vec{E}_L e^{i(k_L z - \omega t)} + \vec{E}_R e^{i(k_R z - \omega t)}$$

$$\vec{E}_L e^{-i\omega t} + \vec{E}_R e^{-i\omega t} = \hat{x} E e^{-i\omega t}$$

$$\vec{E}_L = \xi_L \hat{e}_+ \qquad \vec{E}_R = \xi_R \hat{e}_-$$

$$\hat{e}_+ = \frac{\sqrt{2}}{2}(\hat{x} + i\hat{y}) \qquad \hat{e}_- = \frac{\sqrt{2}}{2}(\hat{x} - i\hat{y})$$

Example: Faraday Rotation

$$\mathcal{E}_L \frac{\sqrt{2}}{2}(\hat{x}+i\hat{y}) + \mathcal{E}_R \frac{\sqrt{2}}{2}(\hat{x}-i\hat{y}) = \hat{x}\,E$$

$$\Rightarrow \quad \mathcal{E}_L = \mathcal{E}_R = \frac{\sqrt{2}}{2}E$$

$$\vec{E}(\vec{x},t) = \frac{\sqrt{2}}{2}E\hat{e}_+ e^{i(k_L z-\omega t)} + \frac{\sqrt{2}}{2}E\hat{e}_- e^{i(k_R z-\omega t)}$$

$$\Rightarrow \quad E_x(\vec{x},t) = \frac{1}{2}iE\left(e^{i(k_L z-\omega t)} + e^{i(k_R z-\omega t)}\right) = 0$$

Example: Faraday Rotation

$$\Rightarrow \quad e^{i(k_L z - \omega t)} = -e^{i(k_R z - \omega t)}$$

$$k_L z - \omega t - (k_R z - \omega t) = (2m+1)\pi$$

m is an integer

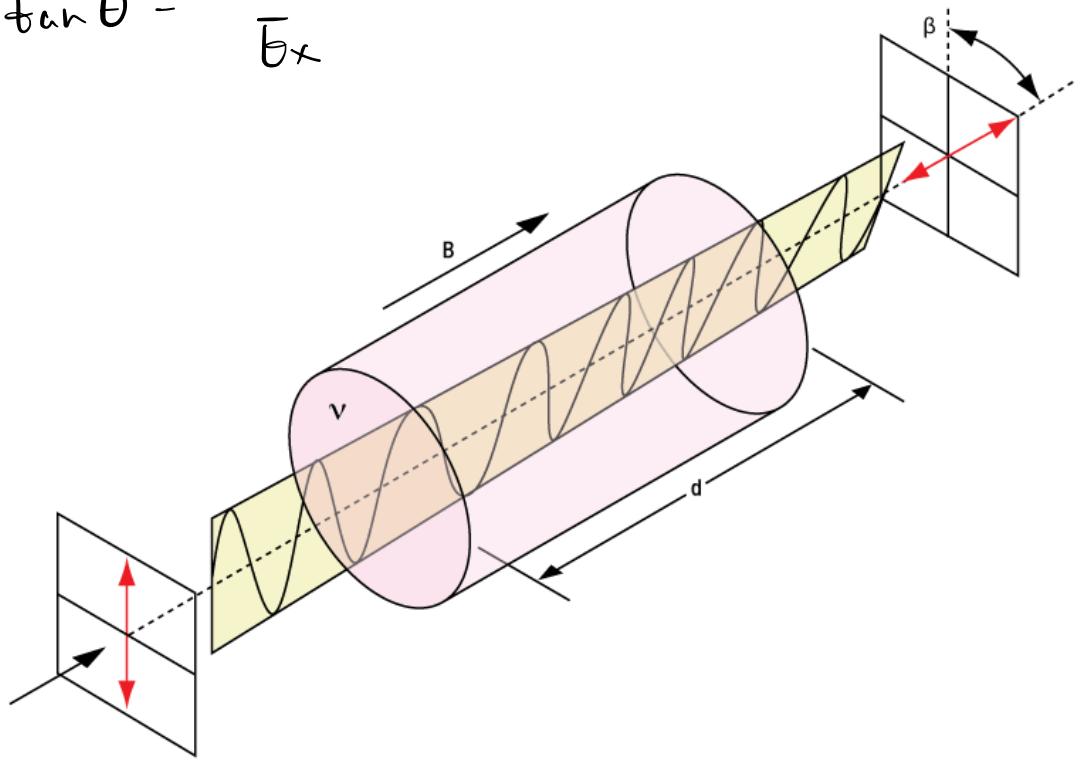
$$\Rightarrow \quad k_L z - k_R z = (2m+1)\pi$$

$$z = \frac{(2m+1)\pi}{k_L - k_R} = \frac{(2m+1)\pi}{\frac{\omega}{c}(n_L - n_R)}$$

$$k = \frac{\omega n}{c}$$

Application of Faraday Rotation

$\tan\theta = \frac{E_y}{E_x}$ $\theta(z)$? HW 1.



Reflection and transmission of plane wave

Assume constant, real, positive ϵ and μ .

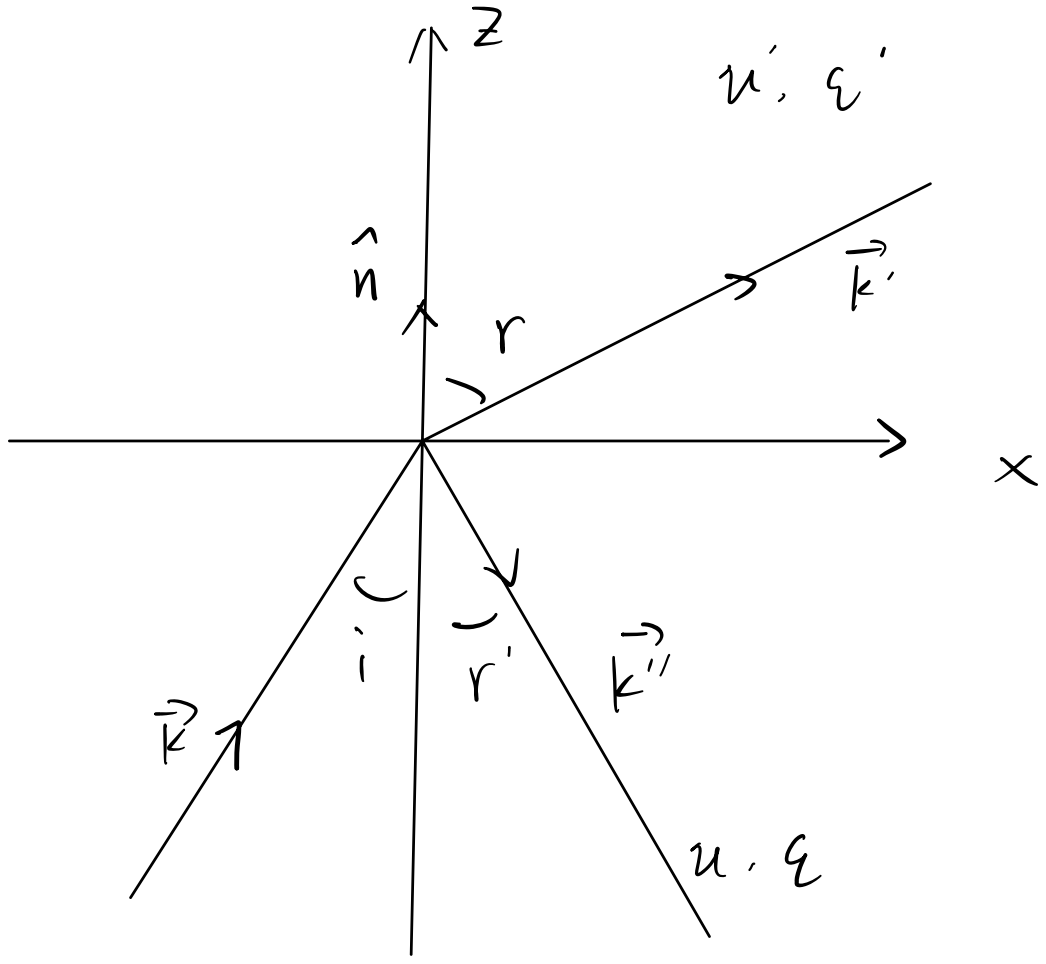
Kinematic properties:

- Angles of reflection and refraction.

Dynamic properties:

- Intensities of reflected and refracted radiation.
- Phase changes and polarization.

Reflection and transmission



Reflection and transmission

- Incident waves:

$$\vec{E}(\vec{x},t) = \vec{E}_0 e^{i(\vec{k}\cdot\vec{x} - \omega t)}$$

$$\vec{B}(\vec{x},t) = \sqrt{\mu\epsilon}\,\frac{\vec{k}\times\vec{E}}{k}$$

Known

- Transmitted waves:

$$\vec{E}'(\vec{x},t) = \vec{E}_0' e^{i(\vec{k}'\cdot\vec{x} - \omega' t)}$$

$$\vec{B}'(\vec{x},t) = \sqrt{\mu'\epsilon'}\,\frac{\vec{k}'\times\vec{E}'}{k'}$$

Unknown

Reflection and transmission

- Reflected waves

$$\vec{E}''(\vec{x},t) = \vec{E}_0'' e^{i(\vec{k}''\cdot\vec{x} - \omega''t)}$$

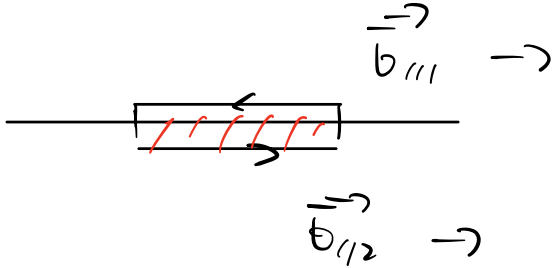
$$\vec{B}''(\vec{x},t) = \sqrt{\mu\varepsilon}\,\frac{\vec{k}''\times\vec{E}''}{k''}$$

- Magnitude of k

$$k = \omega\sqrt{\mu\varepsilon} \qquad k' = \omega'\sqrt{\mu'\varepsilon'} \qquad k'' = \omega''\sqrt{\mu\varepsilon}$$

$$\vec{E}_0',\ \vec{E}_0'',\ \vec{k}',\ \vec{k}'',\ \omega',\ \omega'' \quad ?$$

Reflection and transmission

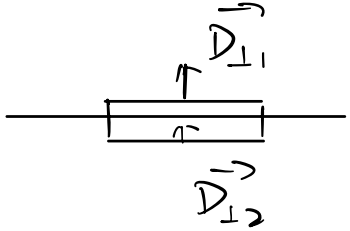


- Boundary condition

$$\int \nabla \times \vec{E}\, ds = \int -\frac{\partial \vec{B}}{\partial t}\, ds$$

$$\begin{cases} \nabla \cdot \vec{D} = 0 \\ \nabla \times \vec{E} = -\dfrac{\partial \vec{B}}{\partial t} \\ \nabla \cdot \vec{B} = 0 \\ \nabla \times \vec{H} = \dfrac{\partial \vec{D}}{\partial t} \end{cases} \Rightarrow \begin{cases} \vec{D}_\perp + \vec{D}_\perp'' = \vec{D}_\perp' \\ \vec{E}_{//} + \vec{E}_{//}'' = \vec{E}_{//}' \\ \vec{B}_\perp + \vec{B}_\perp'' = \vec{B}_\perp' \\ \vec{H}_{//} + \vec{H}_{//}'' = \vec{H}_{//}' \end{cases} \qquad Z = 0$$

$$\nabla \cdot \vec{D} = 0 \qquad \int \vec{D} \cdot d\vec{a} = 0$$



Reflection and transmission

- Phase $\vec{D}_\perp + \vec{D}''_\perp = \vec{D}'_\perp$

$\Rightarrow \varepsilon \vec{E}_\perp + \varepsilon'' \vec{E}''_\perp = \varepsilon' \vec{E}'_\perp \Rightarrow \vec{E}, \vec{E}', \vec{E}''$ must be in phase

$\Rightarrow e^{i(\vec{k}\cdot\vec{x} - \omega t)} = e^{i(\vec{k}''\cdot\vec{x} - \omega' t)} = e^{i(\vec{k}'\cdot\vec{x} - \omega' t)}$

$\Rightarrow \vec{k}\cdot\vec{x} - \omega t = \vec{k}''\cdot\vec{x} - \omega' t = \vec{k}'\cdot\vec{x} - \omega' t$

$\Rightarrow \omega = \omega' = \omega''$ $\quad \vec{k}\cdot\vec{x} = \vec{k}''\cdot\vec{x} = \vec{k}'\cdot\vec{x}$

Reflection and transmission

$$k \sin i = k'' \sin r' = k' \sin r$$

$$k'' = \omega \sqrt{\mu' \varepsilon'} = \omega \sqrt{\mu \varepsilon} = k$$

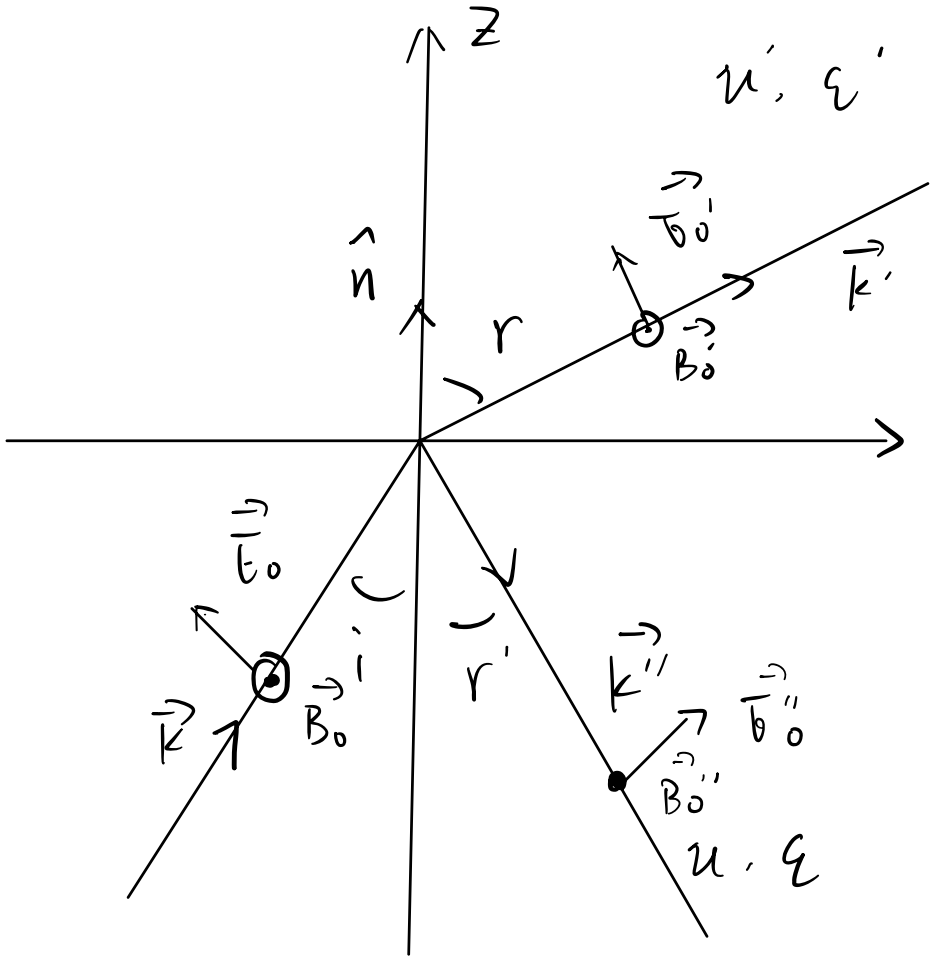
$$k' = \omega \sqrt{\mu' \varepsilon'}$$

$$\sin i = \sin r' = \sin r \sqrt{\frac{\mu' \varepsilon'}{\mu \varepsilon}}$$

$$\Rightarrow \quad r' = i, \qquad \sin i = \sin r \sqrt{\frac{\mu' \varepsilon'}{\mu \varepsilon}}$$

$$n \sin i = n' \sin r \qquad \text{Snell's Law}$$

E in the plane of k and n (p polarized)



×

$\vec{E}_0$ is in the incident plane. C includes $\vec{k}, \vec{k}', \vec{k}''$

S polarization

$\vec{E}_0$ is perp to the incident plane.

E in the plane of k and n (p polarized)

$$
\begin{cases}
\vec{D}_{\perp} + \vec{D}''_{\perp} = \vec{D}'_{\perp} \\
\vec{E}_{//} + \vec{E}''_{//} = \vec{E}'_{//} \\
\vec{B}_{\perp} + \vec{B}''_{\perp} = \vec{B}'_{\perp} \\
\vec{H}_{//} + \vec{H}''_{//} = \vec{H}'_{//}
\end{cases}
\Rightarrow
\begin{cases}
\left[\varepsilon(\vec{E}_0 + \vec{E}''_0) - \varepsilon' \vec{E}'_0\right] \cdot \vec{n} = 0 \\
(\vec{E}_0 + \vec{E}''_0 - \vec{E}'_0) \times \vec{n} = 0 \\
\left[\vec{k} \times \vec{E}_0 + \vec{k}'' \times \vec{E}''_0 - \vec{k}' \times \vec{E}'_0\right] \cdot \vec{n} = 0 \\
\left[\frac{1}{\mu}(\vec{k} \times \vec{E}_0 + \vec{k}'' \times \vec{E}''_0) - \frac{1}{\mu'}(\vec{k}' \times \vec{E}'_0)\right] \times \vec{n} = 0
\end{cases}
$$